FINAL YEAR PROJECT — 2026

A.U.R.A.

Air-gapped Unified Responsive Agent

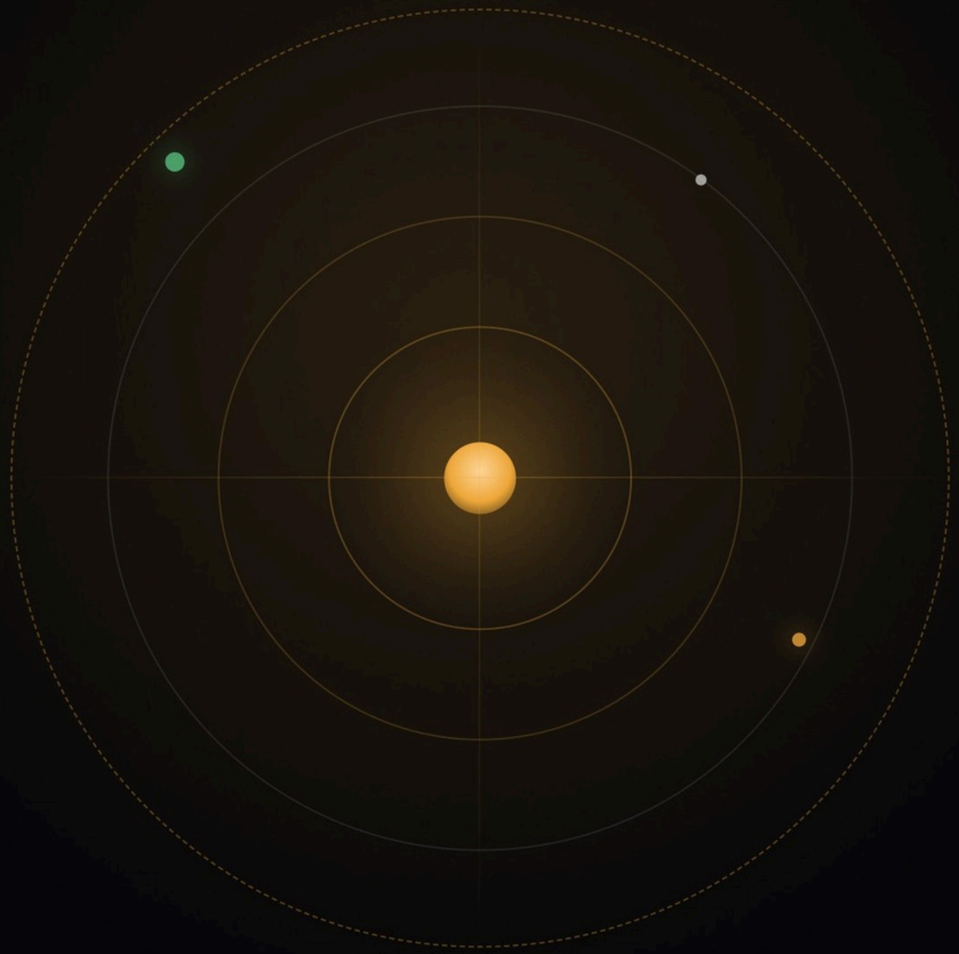
A Locally-Run Multi-Agent System for Secure Computer & OS Automation

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AIR-GAP BOUNDARY · NO OUTBOUND INFERENCE

Project Domains & Nature

01

Artificial Intelligence

Local LLMs & Multi-Agent Reasoning

- Task interpretation
- Planning and decomposition
- Local model inference
- Agent coordination

02

Computer Automation

Operating System & Application Interaction

- Accessibility-based UI interaction
- Background task execution
- File and application operations
- Human-like interaction fallback where required

03

Cybersecurity

Privacy, Access Control & Agent Safety

- Zero-trust execution model
- Prompt-injection defense
- Restricted permissions
- Human approval for high-risk actions

04

Edge Computing

Offline & Hardware-Assisted Computing

- Local inference
- CPU-efficient quantized models
- Raspberry Pi companion node
- Local encrypted state and memory

Technical Problem Statement

Limitations of existing AI assistants

Cloud Dependency

Most intelligent assistants depend on remote APIs and cloud infrastructure. User commands and sensitive contextual data may leave the local machine.

Limited OS Control

Conventional assistants mainly respond through chat or predefined integrations. They cannot reliably perform complex multi-step desktop operations.

Weak Context Persistence

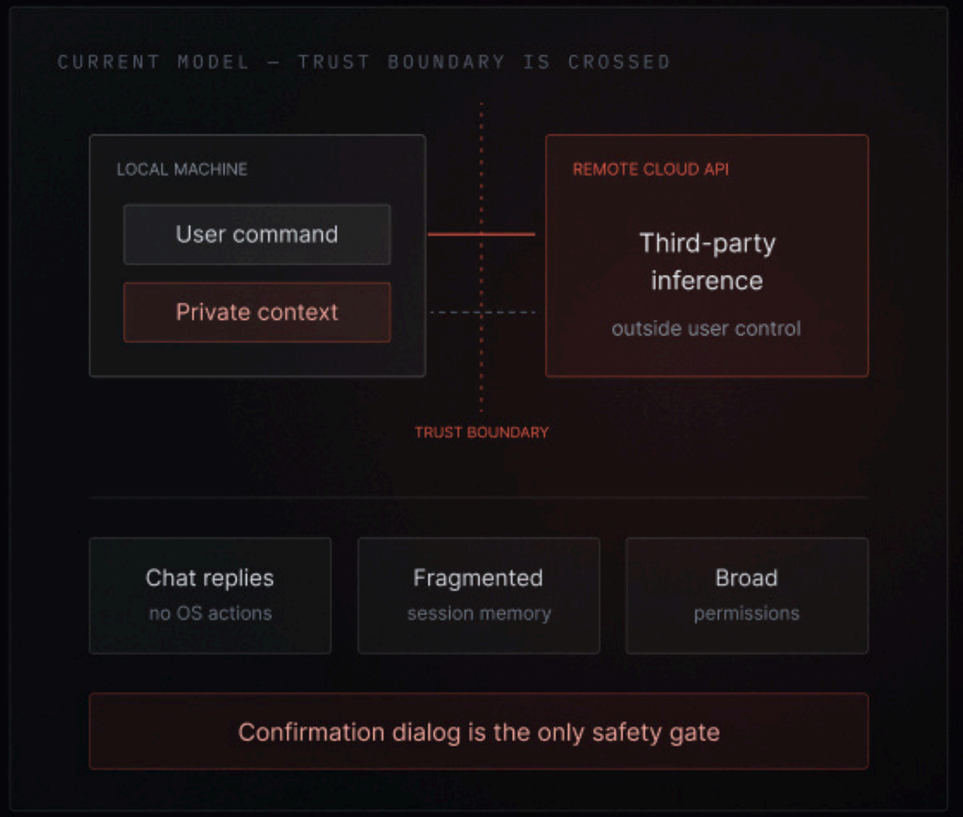
User preferences, habits and previous task context are often fragmented across sessions.

Security Risks

Tool-using AI agents can be exposed to malicious web content, prompt injection and excessive permissions.

No Safe Human Approval Layer

High-impact actions may require stronger authorization than a simple confirmation dialog.



A.U.R.A. — Proposed Solution

01

Local-First Intelligence

Runs reasoning locally using quantized language models without requiring cloud APIs.

02

Multi-Agent Architecture

A Brain → Manager → Specialized Sub-Agent structure divides complex tasks into controlled operations.

03

Direct OS Interaction

Uses accessibility APIs to target application elements directly instead of relying only on visual clicking.

04

Vision Fallback

A local vision model can locate UI elements when accessibility information is unavailable.

05

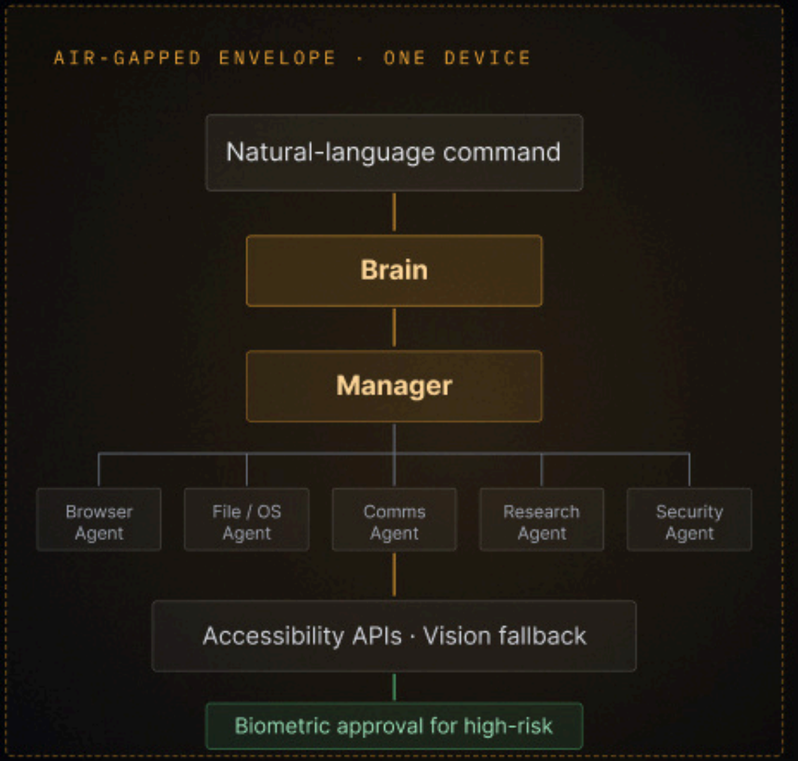
Security by Design

Web content is sanitized before reaching the AI and agents operate within restricted permissions.

06

Human-in-the-Loop Protection

High-risk actions require explicit physical biometric approval before execution.



> Local intelligence + controlled automation + security + human oversight

04 — REAL-WORLD PROBLEM

Why Local Computer Automation Matters

The modern computer user faces three problems

PROBLEM 01

Too Many Repetitive Tasks

- Opening applications
- Managing files
- Searching information
- Moving between applications
- Performing repetitive desktop workflows

PROBLEM 02

Sensitive Data

- Personal documents
- Credentials
- Work files
- Private conversations
- Browser activity

PROBLEM 03

Trust & Control

Users need automation without giving an AI unrestricted access to their entire system.

THE CORE CHALLENGE

Can an AI agent perform useful computer tasks while keeping the user's data, reasoning and execution locally controlled?

A.U.R.A. is designed around this problem.



FOUR CONSTRAINTS, ONE SYSTEM

Conventional Assistant vs. A.U.R.A.

CONVENTIONAL AI ASSISTANT	A.U.R.A.
Cloud-based processing	Local processing
API-dependent intelligence	Local quantized models
Chat-oriented interaction	Task-oriented OS automation
Single assistant workflow	Multi-agent orchestration
Limited persistent context	Hierarchical long-term memory
Accessibility often secondary	Accessibility-first interaction
Vision may be the primary method	Vision used as fallback
Broad permissions can create risk	Restricted-permission execution
Basic confirmation dialogs	HITL biometric approval for high-risk actions
Sensitive state may leave device	Encrypted local storage

A.U.R.A. DESIGN GOAL

Useful automation without sacrificing local privacy and user control.

How A.U.R.A. Works

01
User Command
Natural-language instruction via local interface or voice input

02
Brain
Interprets the user's intent and identifies the overall task

03
Manager
Breaks the task into smaller operations and selects the agents

04
Sub-Agents
Dedicated agents handle specific domains

05
Local Execution
Controlled OS interfaces, accessibility APIs, browser automation, approved tools

06
Verification
Validates the result, records state and reports the outcome to the user

STEP 04 · SPECIALIZED SUB-AGENTS

Browser Agent
Interacts with web applications

File / OS Agent
Locates and manipulates files

Communication Agent
Communication-related tasks

Research Agent
Gathers required information

Security Agent
Validates permissions

Shared context travels with the task across every stage — the Manager keeps agents aligned on one goal.

A.U.R.A. Core Capabilities

01

Multi-Agent Orchestration

- Brain interprets the command
- Manager decomposes the task
- Specialized agents perform domain operations
- Shared context maintained across the workflow



02

Hybrid Local Reasoning

- Local inference without cloud APIs

BitNet b1.58 2B
Lightweight routing and simple decisions

Quantized 7–8B Model
More complex reasoning

03

Background OS Automation

- Accessibility APIs identify application elements
- Direct interaction with UI components
- Tasks can run without repeatedly taking control of the user's foreground window

04

Vision Fallback

- Screenshot-based UI understanding
- Used when accessibility metadata cannot locate an element
- Helps the system adapt to changing application layouts

05

Long-Term Memory

- Stores useful context such as habits, preferences and file references
- Enables continuity between sessions

Security, Control & Hardware

06

Zero-Trust Security

- Untrusted web content is sanitized before reaching the AI
- Restricted execution permissions reduce the impact of compromised tasks

07

Controlled Interaction

- Platform-aware rate limits and interaction policies
- Designed to reduce excessive or unsafe automated activity

08

Biometric HITL Security

- High-risk actions pause execution

Action frozen — Fingerprint verification — Approved

09

Raspberry Pi Companion — the Edge Node



EDGE HARDWARE

Local Processing
on-device, offline

A.U.R.A. Intelligence
agents + reasoning

Secure Response
verified action

- Wake-word microphone
- Fingerprint sensor
- Projector / local feedback

Provides interaction support when the main laptop screen is unavailable.

10

Encrypted Local Storage

- Local state and memory databases are protected using AES-256 encryption

AES-256 · LOCAL ONLY

ChromaDB | SQLite

Encryption layer · keys never leave device

Multi-Agent Task Execution

WORKED EXAMPLE
"Find the project report, summarize it and save the summary in the Documents folder."

STEP 01 — STEP 02 — STEP 03 — STEP 04 — STEP 05 — STEP 06

User Input

Natural-language command

- Typed or spoken instruction
- Received by the local interface
- No content is sent to a cloud API

Intent Understanding

Brain Agent

- User goal
- Required applications
- Required information
- Expected final result

Task Decomposition

Manager Agent

- Locate the report
- Read the relevant content
- Generate summary
- Open target folder
- Save output
- Verify completion

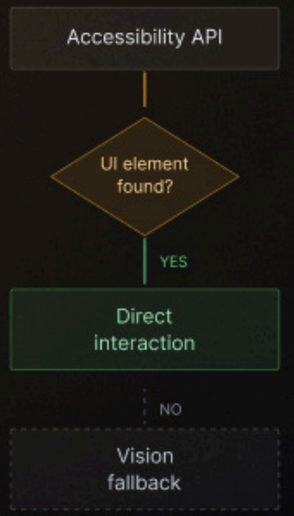
Agent Routing

Domain dispatch

- File / OS — locate and manipulate files
- Research — gather required information
- Browser — interact with web apps
- Communication — handle messaging tasks
- Security — validate permissions

Controlled Execution

Accessibility first



Screenshot-based UI understanding

Result Verification

Close the loop

- Confirm action completion
- Update memory where appropriate
- Record execution state
- Return result to user

Hybrid Quantized AI Architecture

TIER 1 — ROUTING / LIGHTWEIGHT

BitNet b1.58 2B

Designed for

- Fast classification
- Intent routing
- Simple decisions
- Low-resource inference

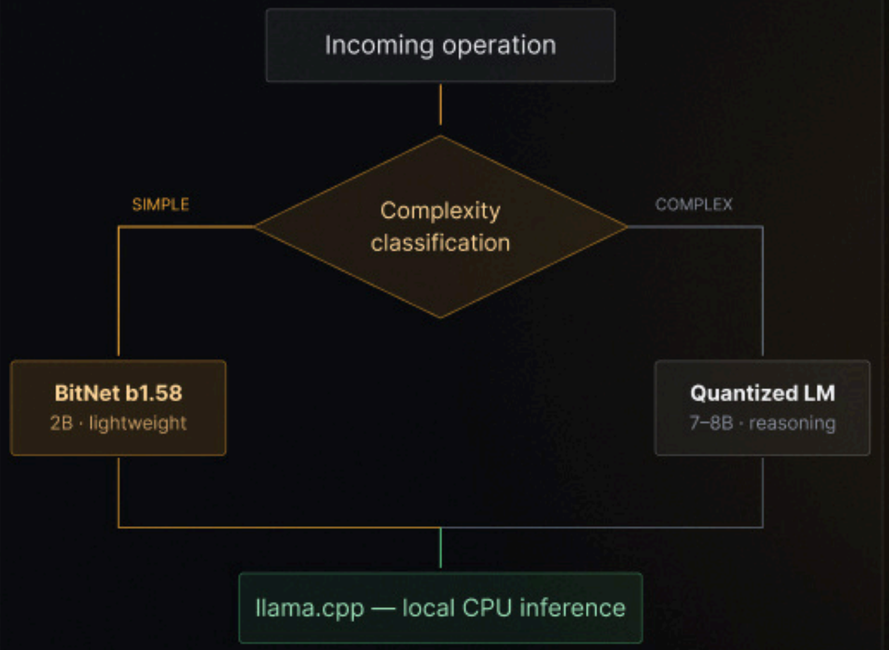
TIER 2 — COMPLEX REASONING

Quantized 7–8B LM

Designed for

- Multi-step reasoning
- Task planning
- Context interpretation
- More complex decision making

WHY HYBRID — ROUTE BY COMPLEXITY



NO OUTBOUND API CALL AT ANY STAGE

EXPECTED BENEFITS

Why not one big model?

- Reduced computational overhead
- Better use of CPU resources
- Lower local inference cost
- More practical offline operation

> The A.U.R.A. proposal specifically defines this two-tier local reasoning approach.

Memory, Security & Agent Control

MODULE 01

Long-Term Memory

Hierarchical memory layer

- User preferences
- Repeated workflows
- Useful file paths
- Relevant historical context



MODULE 02

Security Layer

Zero-trust execution

- Sanitize external web content
- Restrict agent permissions
- Separate untrusted input from trusted instructions
- Prevent unnecessary OS access



MODULE 03

Biometric Control

High-risk action guard

- File deletion
- Sensitive credential access
- Privileged operations

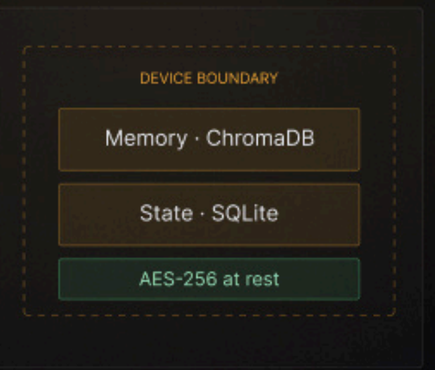


MODULE 04

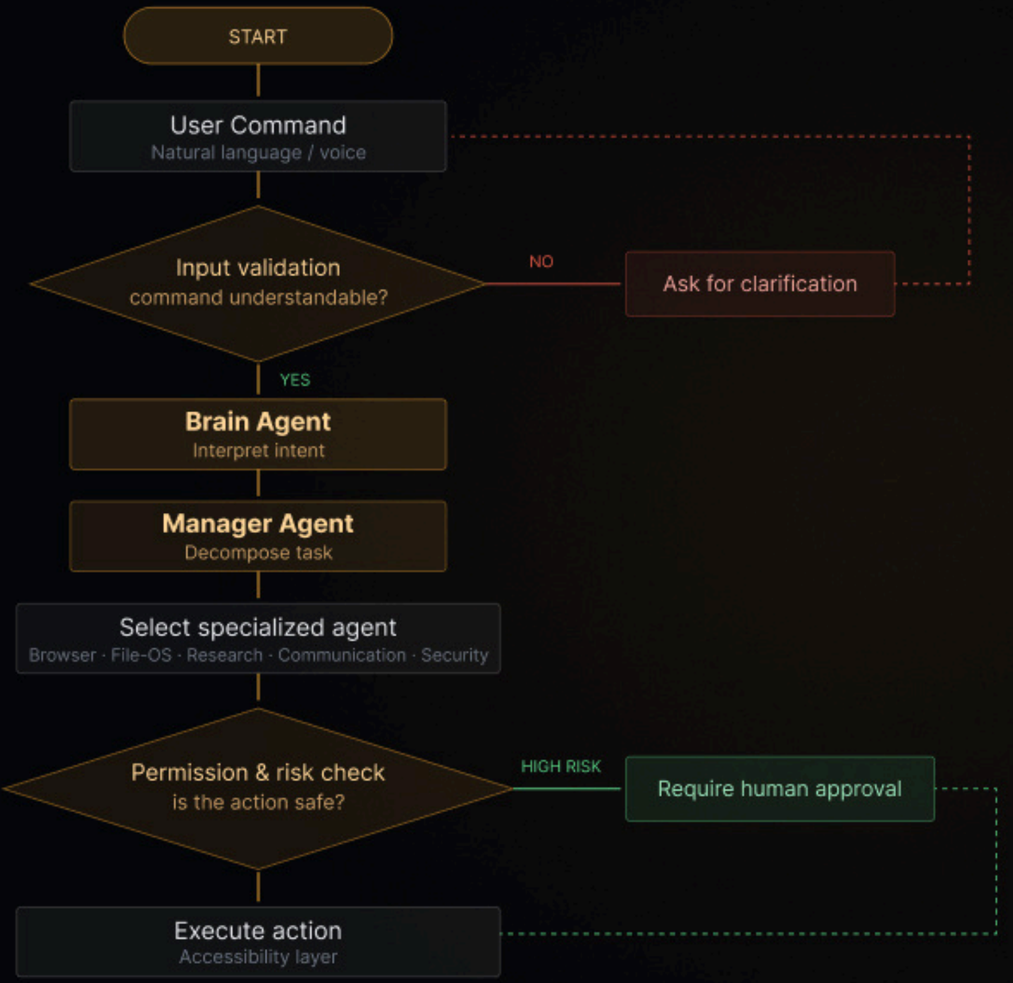
Encrypted State

Local data protection

- ChromaDB
- SQLite
- AES-256 encryption
- Local-only state management



A.U.R.A. End-to-End Process





EXECUTE ACTION CONTINUES →

A.U.R.A. System Architecture

FIVE LAYERS · ONE DEVICE · NO CLOUD DEPENDENCY



Security by Design

01 — LOCAL-FIRST PROCESSING

A.U.R.A. is designed to keep AI inference and user context on the local machine rather than relying on cloud AI APIs.

02 — ZERO-TRUST INPUT HANDLING

External web content is treated as untrusted, reducing the attack surface for indirect prompt injection.

03 — RESTRICTED PERMISSIONS

Agents do not receive unrestricted OS access. Each operation is constrained according to the task and required permissions.

04 — HUMAN-IN-THE-LOOP

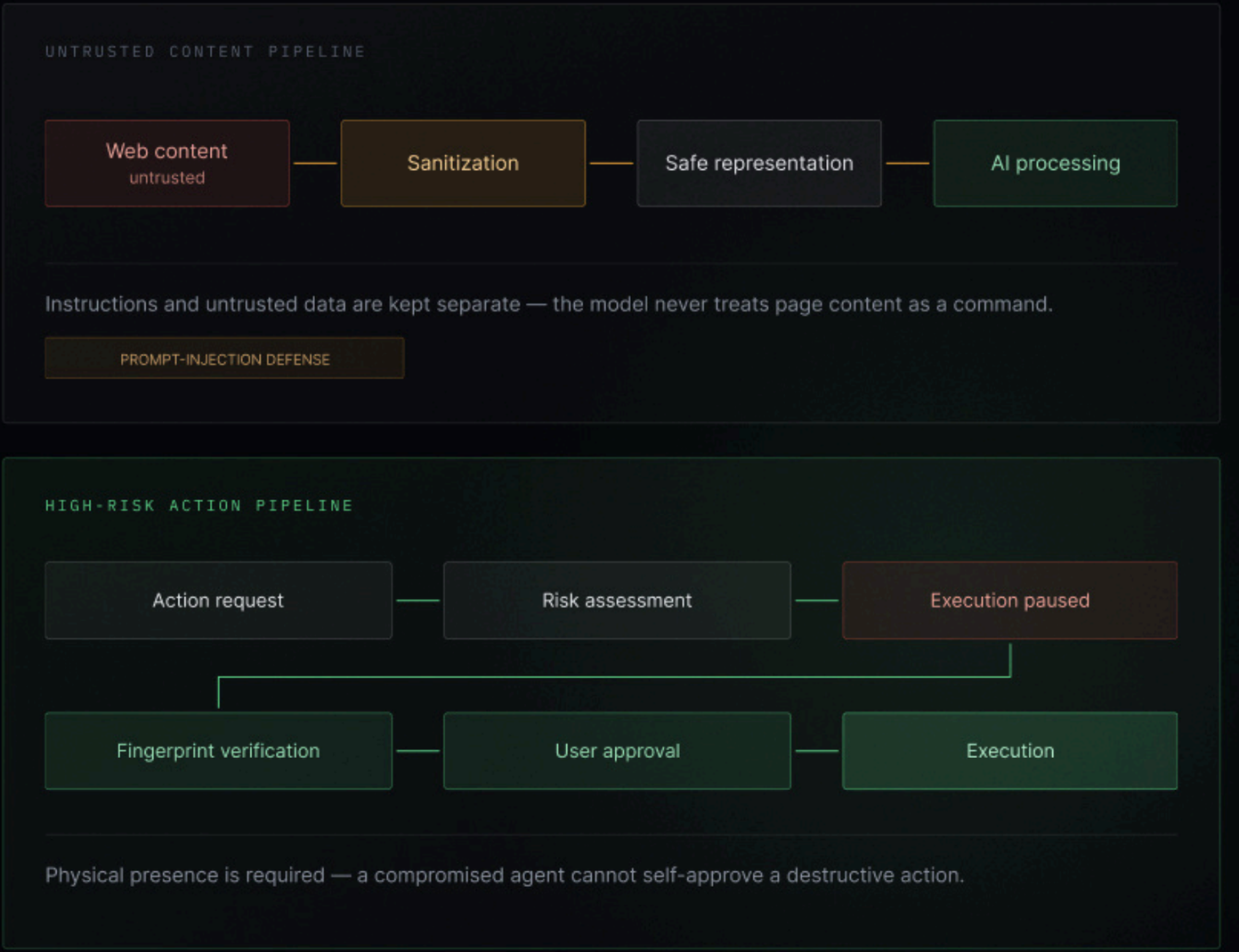
High-risk operations do not execute automatically.

05 — ENCRYPTED LOCAL DATA

Sensitive local state and memory are encrypted using AES-256.

06 — AUDITABLE EXECUTION

Important decisions and actions can be logged so the user can understand what A.U.R.A. attempted and why.



A.U.R.A.

Air-gapped Unified Responsive Agent

A local AI system designed to turn natural-language commands into controlled computer actions.

01

Local Intelligence

AI reasoning can operate locally without cloud AI APIs.

02

Smart Automation

A multi-agent architecture allows complex tasks to be decomposed and handled by specialized agents.

03

Resilient Interaction

Accessibility-based automation is supported by vision fallback when traditional UI information is unavailable.

04

Privacy & Security

User data remains locally controlled through sanitization, restricted permissions, biometric approval and encrypted storage.

THE CORE IDEA

“AI that can act on your computer — while the computer remains under your control.”

Thank You

A.U.R.A. — AIR-GAPPED UNIFIED RESPONSIVE AGENT

Questions & discussion

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